Lecture 1: Flowchart & Pseudocode + Installation

DSA Series | Shradha Khapra (Apna College) | C++ | Short Notes

1. Series ka intro

- DSA = interviews ka main part, problem-solving skill dikhata hai, salary bhi isi se judi hai.
- Topics: basics (variables, data types, operators, if-else, loops, functions) -> DSA -> DP, greedy.
- **Prerequisite:** C++ ki basic understanding.
- **Notes:** shuru ke chapters easy hain, zyada notes nahi. Practice karo. Notes complex topics ke liye.
- Practice questions balanced honge (na zyada, na kam).

2. Problem solve karne ke steps

- **Samjho:** problem kya poochh rahi hai? (e.g. A aur B ka sum)
- **Input:** kya diya hai? (A, B)
- **Logic:** kaise nikalna hai? (A + B)
- Pehle **paper par** solution, phir code. Logic clear = code easy.

3. Flowchart

Solution ka **diagram**: thought process step by step.

Shape	Kaam
Oval	Start / Exit
Parallelogram	Input aur Output (print)
Rectangle	Process (assignment, calculation)
Diamond	Decision (Yes / No condition)
Arrow	Flow ki direction (sab blocks connected)

4. Pseudocode

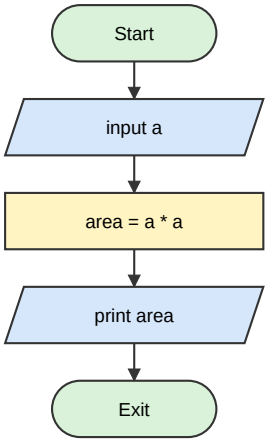
- Logic ko **English jaisi language** mein likhna, kisi bhi developer ko samajh aaye.
- Logic aur code ke beech ka **bridge**.
- Flow: Problem -> Flowchart / Pseudocode -> Code.
- Fixed format nahi, bas logic consistent ho. "Start" likhna zaroori nahi, "Exit" optional.

5. Practice Questions (Flowchart + Pseudocode)

Q1: Square ka area

- Side = a, area = a x a.
- Input -> process -> output.

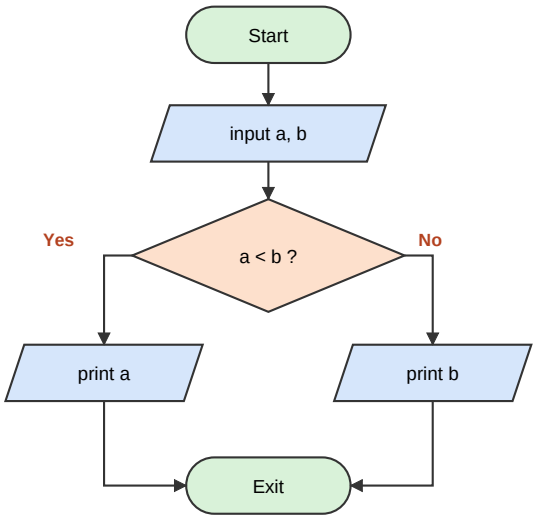
```
input a
area = a * a
print area
exit
```



Q2: Do numbers mein minimum

- Diamond mein compare: a < b ?
- Yes -> print a, No -> print b.
- Example: a=10, b=5 -> 10 < 5 false -> **5**.
- < ya >, dono se logic chalta hai.

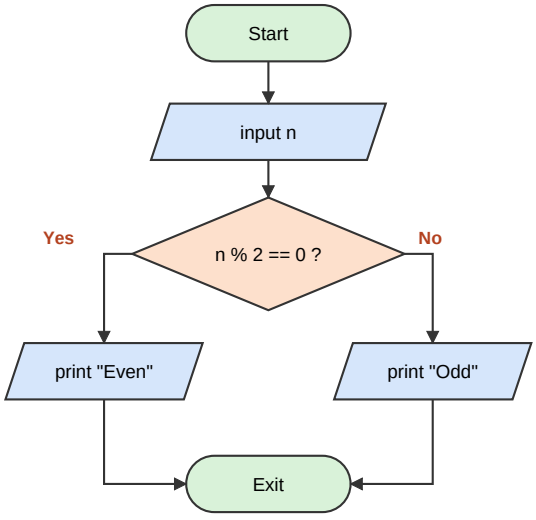
```
input a, b
if a < b
    print a
else
    print b
exit
```



Q3: Even ya Odd

- **%** (modulus) = remainder. 10 % 2 = 0.
- n % 2 == 0 -> Even, warna Odd.
- **==** compare karta hai, **=** assign karta hai.

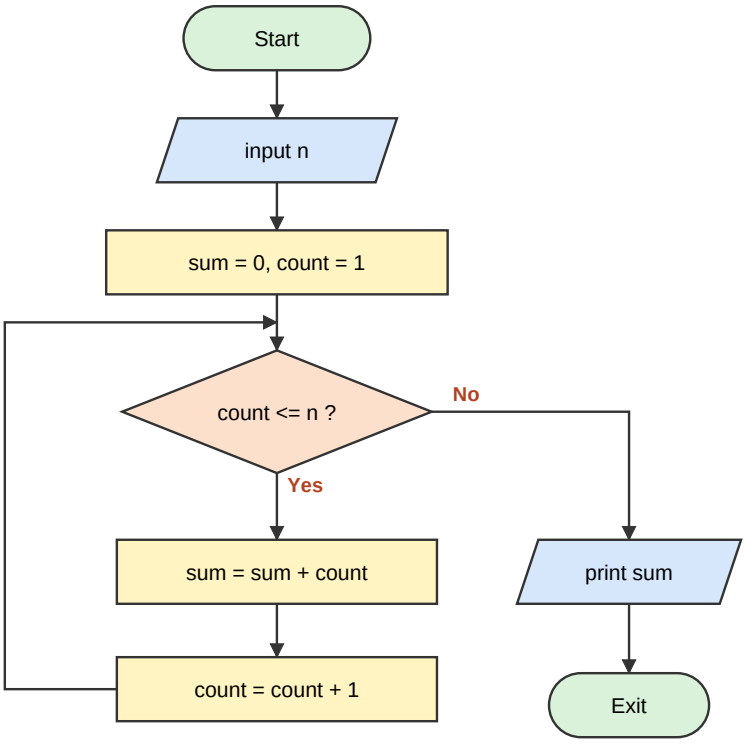
```
input n
if n % 2 == 0
    print "Even"
else
    print "Odd"
```



Q4: 1 se n tak ka sum

- Formula n(n+1)/2 hai, par yahan **logic building** sikhna hai.
- **Initialization:** sum = 0, count = 1.
- Jab tak count <= n: sum mein count jodo, phir count + 1. Yahi **loop** hai.
- Dry run (n = 3): 0+1 = 1, 1+2 = 3, 3+3 = **6**.
- Shorthand: **a += 5** means a = a + 5.

```
input n
sum = 0
count = 1
while count <= n
    sum = sum + count
    count = count + 1
print sum
```

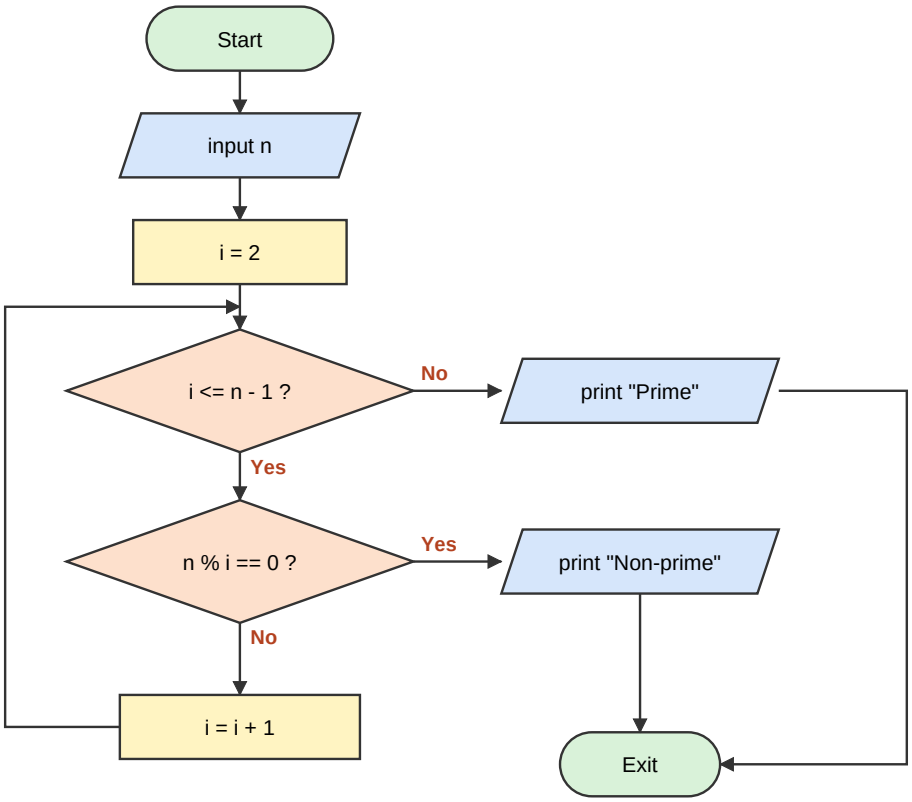


Q5: Prime number check

- **Prime:** sirf 2 factors, 1 aur khud (2, 3, 5, 7, 11, 13). **2** akela even prime hai.
- **Non-prime** = composite (e.g. 4, 6).
- i = 2 se n-1 tak check karo. Kisi i par **n % i == 0** -> Non-prime.
- Koi divisor nahi mila (loop khatam) -> Prime.
- Optimal method nahi, sirf beginners ke liye.
- Do conditions alag hain: **main loop condition** (i <= n-1) aur **andar ki if** (n % i == 0).

```
input n
i = 2
while i <= n - 1
    if n % i == 0
        print "Non-prime"
        exit
    i = i + 1
print "Prime"
exit
```

Extra note: ye logic n = 1 ya 0 ke liye galat ("Prime") bolega, kyunki loop chalta hi nahi. Real code mein pehle check karo: **n <= 1 -> prime nahi**.



6. Homework (khud karo)

- Simple Interest: P x R x T
- Do numbers mein **maximum**
- **Factorial**: n! = 1 x 2 x ... x n
- Driving license eligibility (India: age **>= 18**)
- Prime flowchart ka **n = 8** par dry run